EXECUTED TRANSFER STUDY / 5 NOVEMBER 2026

Frozen PANNs + Linear Classification

ESC-50 development-validation results

91.21% accuracy | 91.10% macro-F1

363 / 398 clips correct. One fixed feature extraction and one linear-head fit on Colab Tesla T4.

Primary paired gain over Channel Attention: +16.83 pp

Conditional 95% source-bootstrap interval: +12.61 to +21.35 pp.

All three new scientific figures and all 50 class results are included. External pretraining, larger encoder capacity and a different frontend distinguish this system from the scratch-trained references. Independent generalization and noise robustness are not established.

A separate research phase following the original 65-task study. Actual run artifacts support the results; ChatGPT roles assisted code drafting and review.

1. Main finding and decision

A frozen AudioSet-pretrained PANNs Cnn14_16k representation with one fixed logistic-regression head classified **363 of 398 validation clips correctly: 91.21% accuracy and 91.10% macro-F1**. No encoder fine-tuning or hyperparameter search was performed.

The primary paired difference against the mean accuracy of the three existing Channel Attention fits was **+16.83 percentage points (pp)**, with a conditional 95% source-bootstrap interval of **[+12.61, +21.35] pp**. This exceeds the execution-time-frozen +3 pp practical screening threshold. That threshold is a project decision rule, not a statistical theorem.

The result supports retaining frozen pretrained audio features plus a linear head as the stronger clean-validation candidate in this project. It does not independently confirm generalization, isolate the effect of pretraining from model capacity or frontend changes, or establish noise robustness. The existing scratch-trained CNN and CRNN remain useful controlled baselines.

2. Question, protocol and evaluation scope

The question was fixed before any new feature extraction or head-fitting result: does this externally pretrained system outperform the existing Channel Attention reference on the same development recordings?

The execution-time-frozen protocol [P01] specifies one embedding extraction, one deterministic head fit, C=1 without search, the reference fits, source resampling and the stopping rule. This is not external preregistration. The broader project has previously exposed every official ESC-50 fold, including the current development-validation fold.

Partition	Role	Clips
Official folds 1-3	Train-only scaler and linear-head fit; exclude sources appearing in validation or test roles	1,200
Official fold 4	Development validation; exclude sources appearing in fold 5	398 from 295 sources
Official fold 5	Metadata used only to exclude source overlap; no waveform decoding or model scoring in this phase	400 designated clips

The dataset revision remains 33c8ce9eb2cf0b1c2f8bcf322eb349b6be34dbb6. All compared validation predictions are aligned by filename and target. Source grouping addresses known recording IDs; perceptual duplicates and AudioSet/ESC-50 pretraining overlap have not been comprehensively excluded.

3. Frozen encoder and linear classifier

The PANNs author repository provides the Cnn14_16k implementation and its official checkpoint. The source revision used here was d2f4b8c18eab44737fcc0de1248ae21eb43f6aa4; the checkpoint identifier is Cnn14_16k_mAP=0.438.pth. Its filename's mAP refers to AudioSet, not this ESC-50 accuracy.

Component	Fixed implementation
Waveform	Mono, scipy resample_poly to 16 kHz; exactly five seconds; no peak normalization
Encoder frontend	Author model's internal Log-Mel: FFT/window 512, hop 160, 64 bands, 50-8,000 Hz
Representation	2,048-dimensional embedding after fc1 ReLU; eval mode disables dropout and augmentation

Component	Fixed implementation
Encoder state	Strict state-dictionary loading; all parameters frozen; inference_mode
Scaling	StandardScaler fitted only on the 1,200 training embeddings
Head	Multiclass logistic regression; C=1, L2 penalty, lbfgs, max_iter=3,000, tol=1e-6
Repetition	One fixed head fit; no nominal seed repetitions, tuning, augmentation or encoder fine-tuning

For one clip, let e be its 2,048-dimensional frozen embedding. Training-set means and scales produce z=(e-mu_train)/sigma_train. The head computes 50 logits Wz+b, then predicts the largest logit. W has shape [50, 2048], and b has shape [50]. Thus the head has 102,450 fitted coefficients and biases. Standardization remains an affine transformation of the embedding.

Only W and b are optimized by the regularized multinomial logistic objective. The encoder is nonlinear but frozen; this is a linear probe of its representation. The loaded author model contains 81,033,231 parameters, with zero trainable encoder parameters. Its external pretraining computation is excluded from this phase's timing. The much larger encoder means that a simple head does not imply a small end-to-end model.

4. Clean-validation comparison

System	Accuracy (%), mean +/- sample SD where repeated	Scope
CNN, scratch-trained	72.61 +/- 0.25	Three existing fixed fits
CNN + BiGRU, scratch-trained	73.87 +/- 1.26	Three existing fixed fits
CRNN + Channel Attention, scratch-trained	74.37 +/- 1.15	Three existing fixed fits
Frozen PANNs + linear head	91.21	One fixed encoder and one head fit

The grey reference systems each summarize three existing fits, not three independent audio datasets. PANNs reports one fit; a three-seed SD is neither estimated nor fabricated. References are individual-fit means, not an ensemble classifier.

Figure 1. System comparison: scratch-trained references and the fixed pretrained system - full-page figure atlas.

5. Paired uncertainty and changes in errors

The primary estimand is the new system's clip accuracy minus the arithmetic mean of the three fixed Channel Attention accuracies. For each clip, the new correctness indicator is compared with the mean of its three reference correctness indicators. The analysis resamples 295 src_file groups with replacement 4,000 times using RNG seed 20261105. Every sampled source retains all constituent clips; each resample divides by its actual clip count. The same draws are shared across systems. The resampled denominator ranged from 360 to 450 clips.

Reference three-fit mean	PANNs difference (pp)	Conditional 95% source interval (pp)
CNN	+18.59	[+13.83, +23.57]
BiGRU	+17.34	[+13.27, +21.68]
Channel Attention	+16.83	[+12.61, +21.35]

These intervals condition on the fixed pretrained encoder, fitted head, reference fits and development dataset. They do not include future training randomness, pretraining uncertainty, historical model-selection bias or independent-dataset uncertainty. They are not simultaneous confidence intervals; CNN and BiGRU comparisons are secondary.

Figure 2. Paired accuracy gains with conditional source-bootstrap intervals - full-page figure atlas.

For each Channel Attention fit, the table distinguishes repaired reference errors from correct reference predictions that PANNs loses:

CA seed	Errors repaired	New errors	Net additional correct
101	78	15	63
202	77	11	66
303	86	14	72

PANNs improves on each reference fit, while also introducing some new errors. Of the 41 clips previously misclassified by CNN, BiGRU, Channel Attention and noise-trained CNN in all three fits, PANNs correctly classifies 26 and still misses 15. This is a descriptive analysis of a previously selected error set, not an independent benchmark.

6. Remaining failures

There are 35 remaining validation errors. Wind and washing_machine each have 4/8 correct predictions; footsteps has 5/8. Twenty-eight classes have every available validation clip classified correctly. Each class contains only seven or eight clips, so neither perfect nor low class scores establish class-wide generalization. No new listening study or label audit was performed.

The complete 50-class table is retained in Appendix A and the row-normalized confusion matrix shows all categories.

Figure 3. All-class row-normalized confusion matrix for the fixed pretrained system - full-page figure atlas.

7. Execution, compatibility repair and delivery verification

Execution used Google Colab Tesla T4 through the official Colab CLI and its SSH transport. Python 3.13.16, PyTorch 2.11.0+cu130, NumPy 2.4.6, scipy 1.16.3, scikit-learn 1.6.1, torchlibrosa 0.1.0 and librosa 0.11.0 were recorded in the run summary [P02].

Measured stage	Wall time
Decode, resample and extract training embeddings	23.38 s
Decode, resample and extract validation embeddings	7.00 s
Fit the single linear head	4.41 s
Main program, including model loading and saving outputs	38.05 s

These times exclude package installation, data and checkpoint downloads, prior compatibility repair, external pretraining and report production. They are not comparable to scratch-training-loop times as total model-development costs. L-BFGS converged after 137 iterations with no recorded convergence warning.

The first attempt stopped before extraction because the legacy official checkpoint contains NumPy metadata that newer PyTorch weights-only loading does not allow by default. The repair used a targeted NumPy data-type allowlist while retaining weights_only=True and strict model loading; the failed attempt [P03] is retained. This compatibility repair changed no model, head hyperparameter or score-selection rule.

Saved normalization and linear coefficients were replayed on all 398 original validation embeddings using the portable NumPy head: **zero prediction changes, 363 correct**. This verifies the exported head on saved features; it is not a second encoder extraction or independent replication. Predictions and probability vectors were downloaded before the Colab T4 session was terminated. No resource upgrade or purchase was made.

8. Interpretation and stopping decision

This fixed pretrained system is a substantially better clean-validation candidate under the present protocol. The observed gain is much larger than the earlier +0.50 pp Channel Attention difference, making pretrained representations a more productive practical direction than another attention-module search in this project.

However, the comparison changes external training data, encoder size, learned representation and frontend together. It cannot identify which change causes the gain. Unknown pretraining overlap and repeated development exposure also prevent a claim of fresh independent generalization. PANNs was not evaluated on the Gaussian-noise or gain conditions in this phase; the previous noise-trained CNN findings do not transfer automatically to it.

The fixed experiment is complete and stops here, with no fine-tuning, alternate checkpoint, regularization sweep or additional seed fits. Any future robustness or independent-data study needs its own stated evaluation scope. The previous 65-task study and its conclusions remain a separate record.

9. Evidence and reproduction

Protocol [P01], paired summary [P04], per-fit paired errors [P05], all fixed reference predictions [P06], per-class comparison [P07], clip predictions [P08], probabilities [P09], run log [P10], and portable-head replay [P11] are retained. Code: single extraction and fit [P12] and offline comparison and plotting [P13]. The model card [P14] describes the small exported head and its required external encoder.

Two existing Chrome ChatGPT roles supplied an implementation draft and protocol/result review. Local integration, actual Colab execution and recorded predictions supplied the experimental evidence. Role agreement is not external human review or independent replication.

Appendix A. All 50 validation classes

PANNs counts concern one fit. CA accuracy is the mean of three fixed fits on the same class clips. It is not based on three times as many independent recordings.

Class	Clips	PANNs correct	PANNs accuracy (%)	CA mean accuracy (%)
dog	8	8	100.0	87.5
rooster	8	8	100.0	91.7
pig	8	8	100.0	58.3
cow	8	8	100.0	87.5
frog	8	8	100.0	83.3
cat	8	8	100.0	62.5
hen	7	6	85.7	61.9
insects	8	7	87.5	62.5
sheep	8	8	100.0	29.2
crow	8	7	87.5	87.5
rain	8	8	100.0	83.3
sea_waves	8	7	87.5	87.5
crackling_fire	8	8	100.0	95.8
crickets	8	8	100.0	66.7
chirping_birds	8	8	100.0	95.8
water_drops	8	6	75.0	62.5
wind	8	4	50.0	25.0
pouring_water	8	8	100.0	91.7
toilet_flush	8	8	100.0	87.5
thunderstorm	8	8	100.0	91.7
crying_baby	8	7	87.5	87.5
sneezing	8	7	87.5	91.7
clapping	8	8	100.0	66.7
breathing	8	8	100.0	83.3
coughing	8	8	100.0	91.7
footsteps	8	5	62.5	66.7
laughing	8	7	87.5	50.0

Class	Clips	PANNs correct	PANNs accuracy (%)	CA mean accuracy (%)
brushing_teeth	8	8	100.0	79.2
snoring	8	8	100.0	87.5
drinking_sipping	8	7	87.5	33.3
door_wood_knock	8	6	75.0	83.3
mouse_click	8	7	87.5	87.5
keyboard_typing	8	8	100.0	79.2
door_wood_creaks	8	7	87.5	33.3
can_opening	8	8	100.0	100.0
washing_machine	8	4	50.0	45.8
vacuum_cleaner	8	7	87.5	87.5
clock_alarm	7	7	100.0	100.0
clock_tick	8	7	87.5	58.3
glass_breaking	8	8	100.0	79.2
helicopter	8	6	75.0	20.8
chainsaw	8	6	75.0	87.5
siren	8	8	100.0	50.0
car_horn	8	8	100.0	70.8
engine	8	7	87.5	62.5
train	8	8	100.0	95.8
church_bells	8	7	87.5	79.2
airplane	8	6	75.0	62.5
fireworks	8	8	100.0	100.0
hand_saw	8	8	100.0	100.0

Project evidence index

The paths below refer to the accompanying ESC-50 project. They remain useful provenance references when the PDF is shared, but the project files are not embedded in this PDF. Public web references in the report are clickable.

[P01] `reports/panns_20261105/protocol.json`

[P02] `reports/panns_20261105/run_summary.json`

[P03] `reports/panns_20261105/checkpoint_loading_attempt1.log`

[P04] `reports/panns_20261105/comparison_summary.json`

[P05] `reports/panns_20261105/paired_errors.csv`

[P06] `reports/panns_20261105/reference_predictions.csv`

[P07] `reports/panns_20261105/per_class_comparison.csv`

[P08] `reports/panns_20261105/validation_predictions.csv`

[P09] `reports/panns_20261105/validation_probabilities.csv`

[P10] `reports/panns_20261105/run.log`

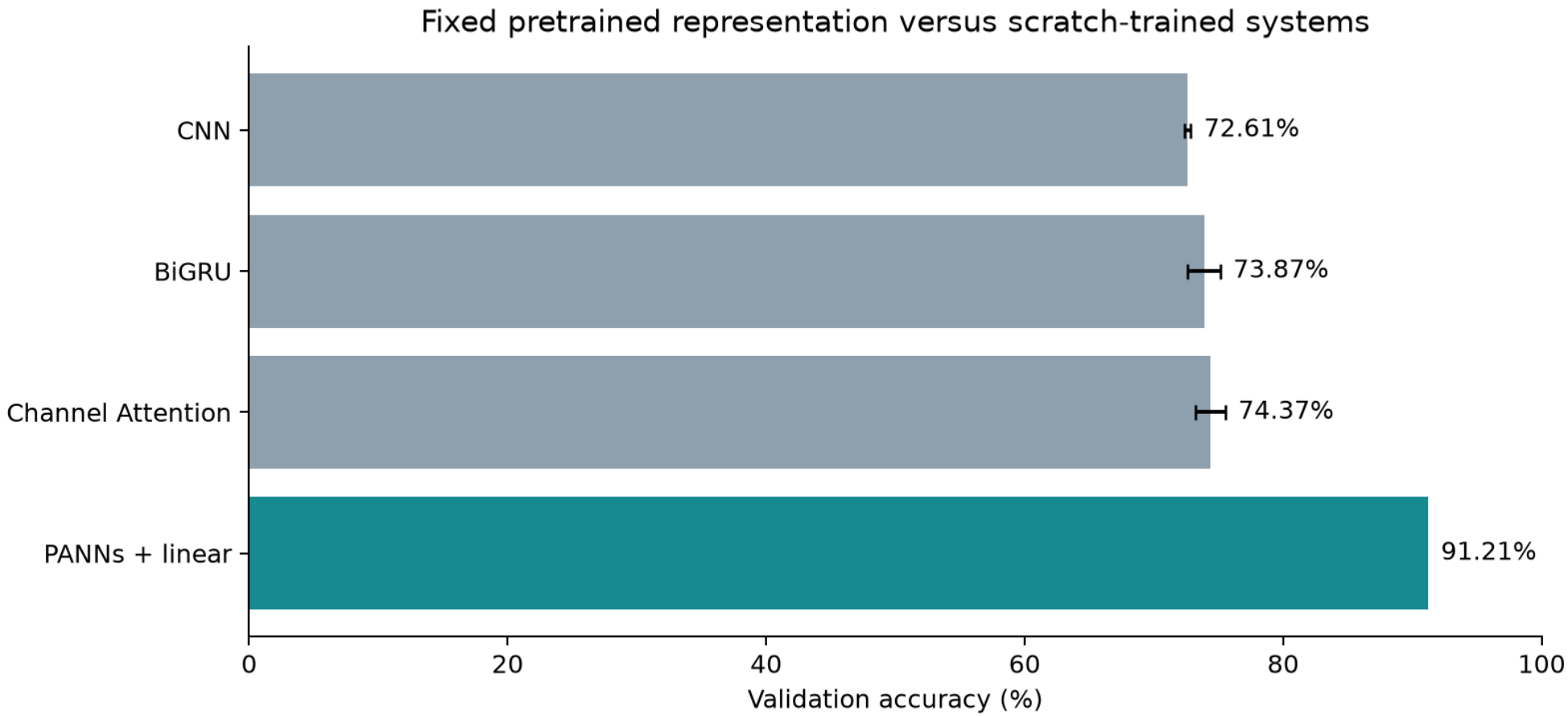
[P11] `reports/panns_20261105/portable_head_replay.json`

[P12] `scripts/run_panns_linear_probe.py`

[P13] `scripts/analyze_panns_transfer.py`

[P14] `models/panns_linear/MODEL_CARD.md`

Figure 1. System comparison: scratch-trained references and the fixed pretrained system

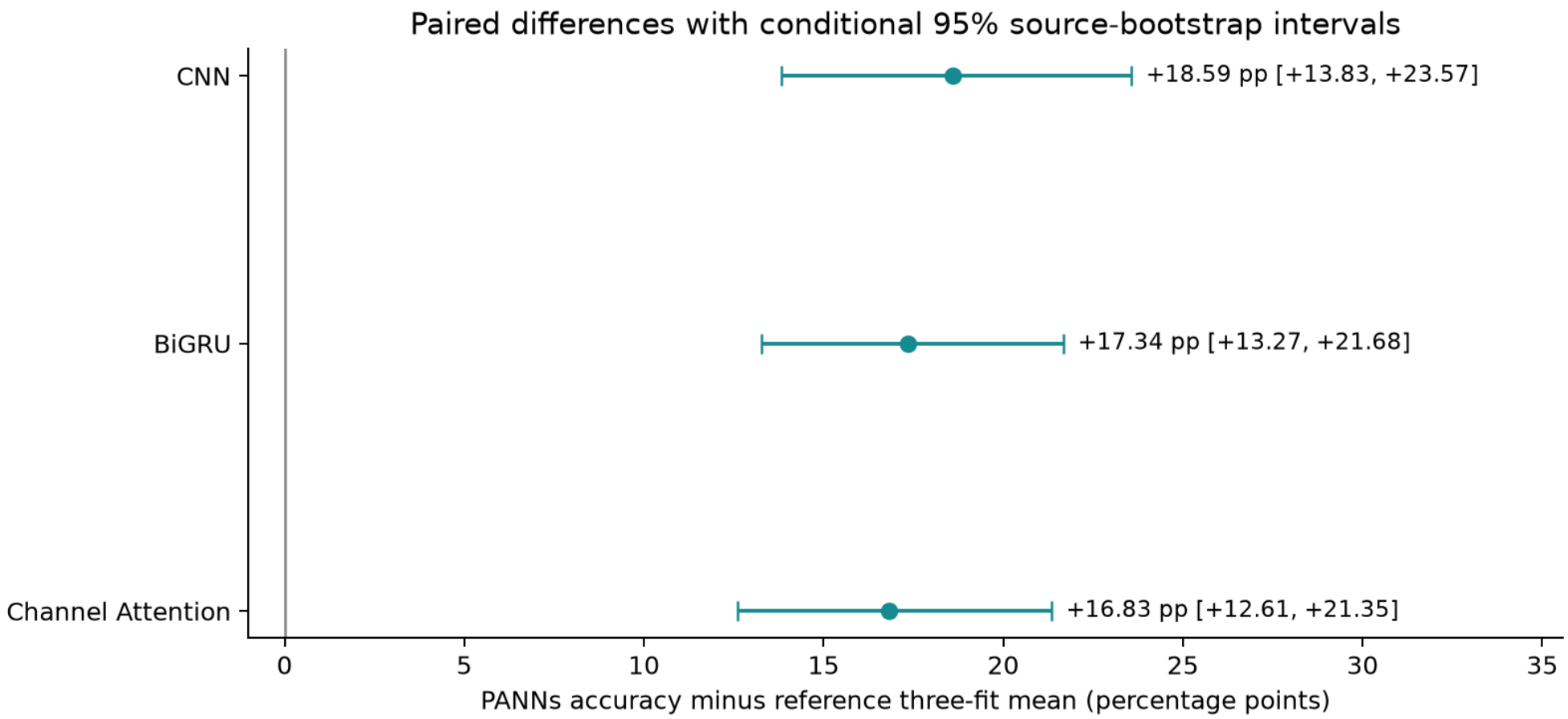


398 development clips / 295 sources. Grey: three-fit mean +/- sample SD. Teal: one deterministic head.
External pretraining, encoder size and frontend differ; no architecture-only causal claim.

System comparison: scratch-trained references and the fixed pretrained system

Source: `reports/panns_20261105/system_comparison.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 2. Paired accuracy gains with conditional source-bootstrap intervals

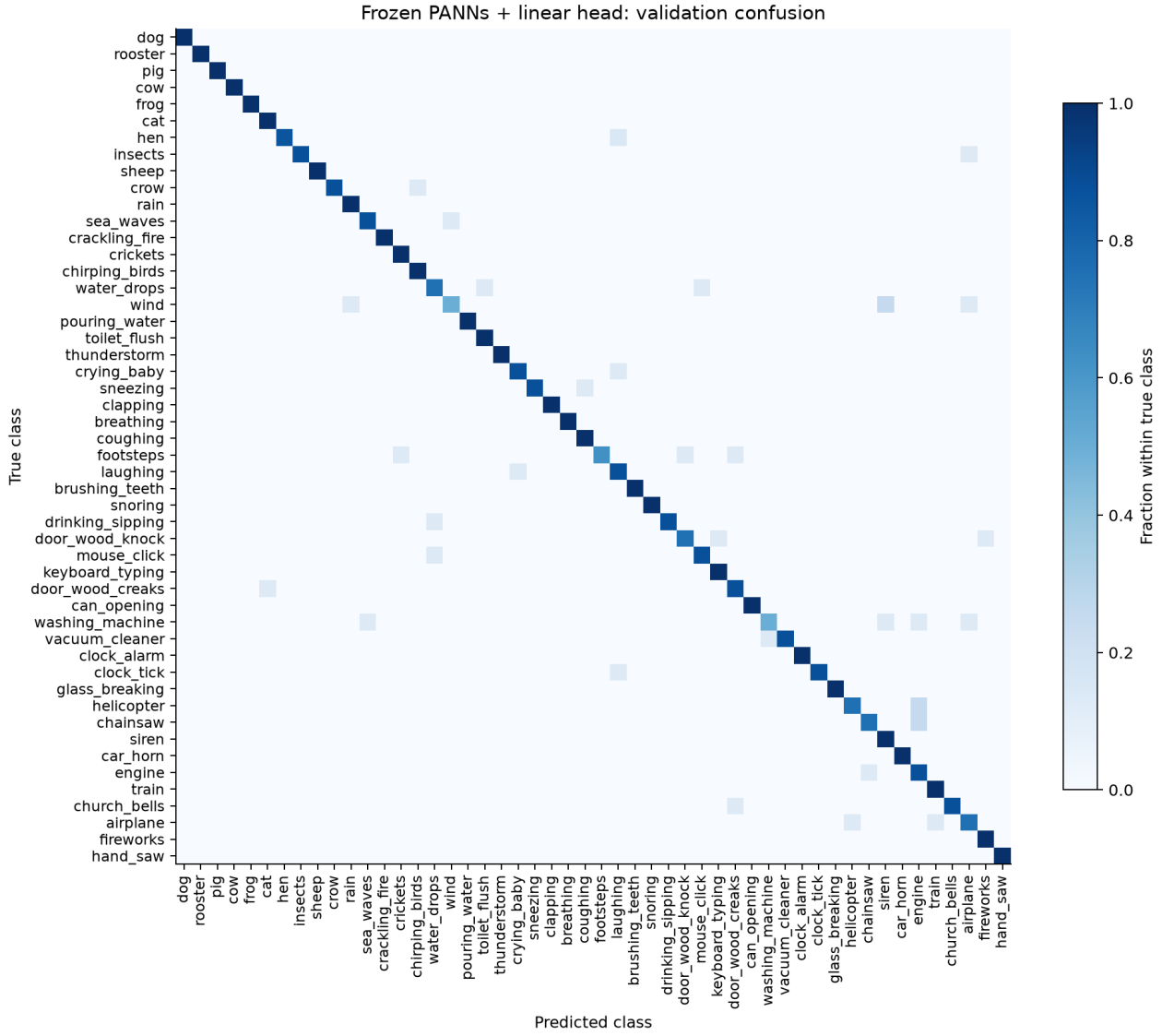


4,000 source-cluster resamples; retain all clips, varying denominator. Fixed fitted models; not independent confirmation.

Paired accuracy gains with conditional source-bootstrap intervals

Source: `reports/panns_20261105/paired_accuracy_gains.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 3. All-class row-normalized confusion matrix for the fixed pretrained system



All-class row-normalized confusion matrix for the fixed pretrained system

Source: `reports/panns_20261105/confusion_matrix.png`. Original saved figure; no experiment was rerun for this PDF.